Module 05 Lab - Data Preparation**Objective:** To learn and apply the most common data preparation techniques. Raw data is rarely ready for a machine learning model. This process, also called preprocessing, is one of the most critical steps in the entire ML workflow.**In this lab, you will write more of the code.** Read the explanations and then complete the tasks in the code cells.

Part 1: Setup and Initial LookWe will continue using the Titanic dataset because it has the exact problems we need to solve: missing values and non-numeric data.

```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/dat

# Let's look at the missing values
print("--- Missing Values Before Cleaning ---")
print(df.isnull().sum())
```

```
--- Missing Values Before Cleaning ---
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
```

```
Embarked         2
dtype: int64
```

Part 2: Handling Missing Values (Imputation)**Concept:** Most machine learning models cannot handle missing values (`NaN`). We must deal with them. Dropping the rows is an option, but you lose data. A better way is **imputation**, which means filling in the missing values with a calculated guess.Common imputation strategies:* **Mean:** Fill with the average value. Good for normally distributed data.* **Median:** Fill with the middle value. Better for skewed data or data with outliers (like `Fare`).* **Mode:** Fill with the most frequent value. Used for categorical data.

Task 1: Impute the 'Age' ColumnThe 'Age' column is missing many values. Since age can be skewed (e.g., by a few very old passengers), using the **median** is a robust choice.**Your Task:** Calculate the median of the 'Age' column and use the `.fillna()` method to replace the missing values.

```
# Calculate the median age and use it to replace missing Age values
median_age = df['Age'].median()
df['Age'] = df['Age'].fillna(median_age)

# Verify that Age no longer contains missing values
print("Median age used for imputation:", median_age)
print("Missing values in 'Age' after imputation:")
print(df['Age'].isnull().sum())
```

```
Median age used for imputation: 28.0
Missing values in 'Age' after imputation:
0
```

Part 3: Encoding Categorical Features**Concept:** Machine

learning models are mathematical, so they need numbers, not text. We need to convert categorical columns (like 'Sex' and 'Embarked') into a numerical format. The most common method is **One-Hot Encoding**.One-Hot Encoding takes a column with `N` categories and turns it into `N` new columns, each with a `1` or `0`. For example, the 'Sex' column (`male`, `female`) becomes two new columns: `Sex_male` and `Sex_female`.Pandas has a convenient function called `pd.get_dummies()` that does this for us.

Task 2: One-Hot Encode Categorical Columns**Your Task:** Use `pd.get_dummies()` to encode the 'Sex' and 'Embarked' columns. Make sure to drop the original columns after encoding.

```
# One-hot encode Sex and Embarked and remove the original categorical
encoded_df = pd.get_dummies(df, columns=['Sex', 'Embarked'], drop_firs

# Display the transformed DataFrame
print(encoded_df.head())
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name   Age  SibSp  Par
0                            Braund, Mr. Owen Harris  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  38.0      1
2                             Heikkinen, Miss. Laina  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  35.0      1
4                           Allen, Mr. William Henry  35.0      0

             Ticket     Fare Cabin  Sex_male  Embarked_Q  Embarked_S
0         A/5 21171   7.2500   NaN      True       False        True
1          PC 17599  71.2833   C85     False       False       False
2  STON/O2. 3101282   7.9250   NaN     False       False        True
3            113803  53.1000  C123     False       False        True
```

```
4           373450   8.0500   NaN      True       False        True
```

Part 4: Feature Scaling**Concept:** Many models are sensitive to the scale of the features. For example, `Age` (from 0-80) and `Fare` (from 0-512) are on very different scales. This can cause the model to incorrectly believe that `Fare` is a more important feature simply because its values are larger.**Feature Scaling** solves this by putting all features on a similar scale. A common method is **Standardization** (`StandardScaler` in scikit-learn), which rescales the data to have a mean of 0 and a standard deviation of 1.**Important:** You only scale your numerical features, not your target variable or your newly encoded categorical columns.

Task 3: Scale the 'Age' and 'Fare' Columns**Your Task:** Use `StandardScaler` from `sklearn.preprocessing` to scale the 'Age' and 'Fare' columns.

```
from sklearn.preprocessing import StandardScaler

# Create the scaler and select the numerical columns to standardize
scaler = StandardScaler()
columns_to_scale = ['Age', 'Fare']

# Fit the scaler and transform Age and Fare
encoded_df[columns_to_scale] = scaler.fit_transform(encoded_df[columns

# Display the scaled data
print(encoded_df.head())
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
```

```
3            4         1       1
4            5         0       3

                                                Name       Age  SibSp
0                            Braund, Mr. Owen Harris -0.565736      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  0.663861      1
2                             Heikkinen, Miss. Laina -0.258337      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  0.433312      1
4                           Allen, Mr. William Henry  0.433312      0

             Ticket      Fare Cabin  Sex_male  Embarked_Q  Embarked_S
0         A/5 21171 -0.502445   NaN      True       False        True
1          PC 17599  0.786845   C85     False       False       False
2  STON/O2. 3101282 -0.488854   NaN     False       False        True
3            113803  0.420730  C123     False       False        True
4            373450 -0.486337   NaN      True       False        True
```

📝Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. **Why is it often better to impute missing values with the median instead of the mean?**
   The median is less affected by unusually high or low values. If the data is skewed or contains outliers, the median can represent a typical value better than the mean.
2. **Explain in your own words what One-Hot Encoding does and why it is necessary.**
   One-Hot Encoding converts categories written as text into numerical columns that a machine learning model can use. Each resulting column indicates whether a row belongs to a particular category using values such as 0 and 1.
3. **Would you need to apply Feature Scaling to a Decision Tree model? Why or why not?**
   Feature scaling is generally not necessary for a Decision Tree. A Decision Tree makes decisions by comparing feature values to split points, so a feature having larger numerical values does not automatically give it more influence over the tree.